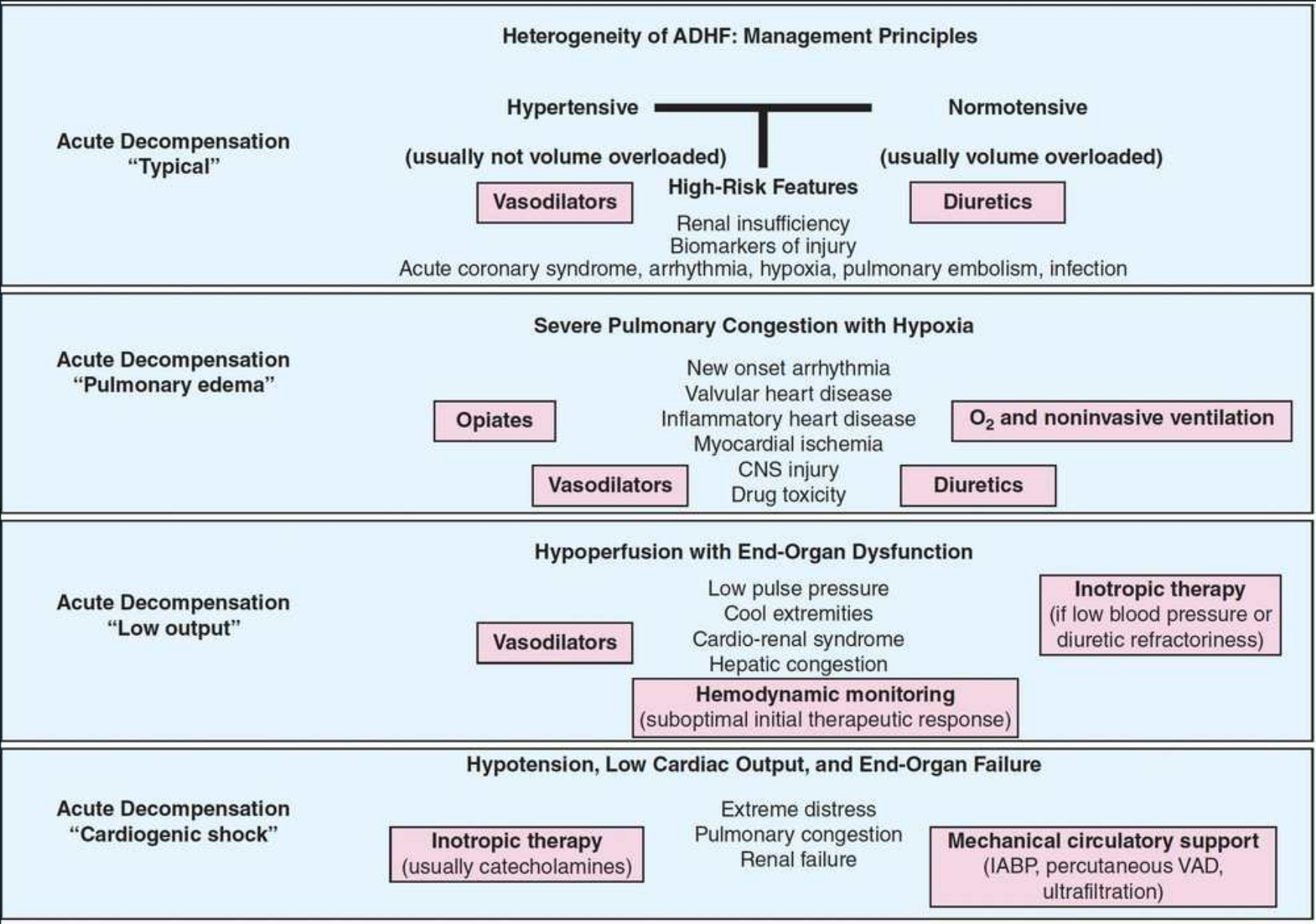


Heart Failure — Figure 265-2: ADHF Phenotypes & Therapy



1. Typical — hypertensive

WHAT YOU SEE

Top band, hypertensive branch

WHAT IT ENCODES

The 'typical' ADHF presentation splits by blood pressure. The HYPERTENSIVE subset (SBP often >140) is usually NOT globally volume-overloaded — it is a problem of acute afterload mismatch and fluid REDISTRIBUTION into the lungs ('vascular failure'/flash pulmonary edema), often with preserved EF. Because total body water may be near-normal, the priority is rapid AFTERLOAD/preload reduction with VASODILATORS rather than aggressive diuresis.

BOARD TRAP

WRONG: every hypertensive ADHF patient is fluid-overloaded and just needs big doses of diuretic. Discriminator — hypertensive 'flash' pulmonary edema is largely fluid REDISTRIBUTION with near-normal total body volume; the key therapy is VASODILATORS (afterload reduction), and over-diuresing these patients risks hypotension and renal injury.

2. Typical — normotensive

WHAT YOU SEE

Normotensive branch → diuretics

WHAT IT ENCODES

The NORMOTENSIVE 'typical' patient has the classic gradual congestion picture — true volume overload accumulated over days to weeks (weight gain, rising JVP, edema, orthopnea). The mainstay is DIURESIS to remove excess sodium and water. This is the prototypical WET-WARM profile (congested but adequately perfused) — the most common ADHF presentation and the one that does best with loop diuretics ± vasodilators.

BOARD TRAP

WRONG: a normotensive, well-perfused, congested patient needs inotropes to improve output. Discriminator — the wet-WARM patient is perfusing fine; treat congestion with DIURETICS, not inotropes (which add no benefit and increase O2 demand/arrhythmia/mortality when perfusion is adequate).

3. Vasodilators (typical)

WHAT YOU SEE

The pink 'Vasodilators' box on the LEFT of the top 'Typical' band, sitting under the Hypertensive branch — flagging vasodilators as the therapy aimed at the high-BP, redistribution phenotype.

WHAT IT ENCODES

Vasodilators (IV nitroglycerin first-line; nitroprusside for severe hypertension or acute valvular regurgitation) are the targeted therapy for hypertensive typical ADHF — they cut preload and afterload, dropping pulmonary capillary wedge pressure within minutes and relieving dyspnea faster than diuretics alone. They REQUIRE adequate blood pressure and are unsafe once the patient is hypotensive.

BOARD TRAP

WRONG: vasodilators can be used across all ADHF phenotypes regardless of blood pressure. Discriminator — vasodilators belong to the HYPERTENSIVE/preserved-BP patient; in the cold/low-output or shock phenotypes they cause harmful hypotension and are avoided.

4. High-risk features

WHAT YOU SEE

Center: renal insufficiency, biomarkers, ACS, arrhythmia, PE, infection

WHAT IT ENCODES

High-risk features that mandate admission/intensified care and a search for triggers: renal insufficiency/worsening renal function, elevated biomarkers of myocardial injury (troponin) and very high natriuretic peptides (BNP >~400–500, NT-proBNP age-adjusted), acute coronary syndrome, new arrhythmia (AF with RVR), hypoxia, pulmonary embolism, and infection/sepsis. Always identify the precipitant — recall mnemonic 'FAILURE / CHAMPS' (forgot meds, arrhythmia, ischemia, infection, etc.). Troponin/BNP elevation portends worse prognosis.

BOARD TRAP

WRONG: in a patient with known chronic HF you don't need to look for a precipitant — just treat the congestion. Discriminator — every ADHF admission requires a hunt for the TRIGGER (ACS, arrhythmia, infection, PE, nonadherence, NSAIDs) and risk stratification with troponin/BNP; missing an MI or PE driving the decompensation is the classic error.

5. Pulmonary edema phenotype

WHAT YOU SEE

Second band: severe pulm congestion with hypoxia

WHAT IT ENCODES

The 'acute pulmonary edema' phenotype is dominated by severe pulmonary congestion with HYPOXIA and marked respiratory distress, frequently of abrupt onset. Common triggers: new arrhythmia, valvular heart disease (acute MR/AS), inflammatory/ischemic heart disease, CNS injury (neurogenic pulmonary edema), and drug toxicity. Management combines oxygenation/ventilatory support, vasodilators, and diuretics, plus correcting the trigger.

BOARD TRAP

WRONG: acute hypoxic pulmonary edema should be managed with fluids and observation until imaging confirms the cause. Discriminator — this is an emergency requiring immediate OXYGEN/NIV, vasodilators, and diuretics; giving fluids worsens the edema, and delaying support to await workup is dangerous.

6. Opiates

WHAT YOU SEE

The pink 'Opiates' box on the left of the second 'Pulmonary edema' band — a preload-reducing adjunct now used cautiously in acute cardiogenic pulmonary edema.

WHAT IT ENCODES

Opiates (e.g., morphine) were traditionally used in acute pulmonary edema to reduce preload (venodilation), blunt the sympathetic surge, and relieve dyspnea/anxiety. Note the modern caveat: observational data link morphine in acute HF to more adverse events (respiratory depression, hypotension, increased intubation/mortality signals), so use is now cautious and selective rather than routine.

BOARD TRAP

WRONG: morphine is a standard, benign first-line drug for everyone in acute cardiogenic pulmonary edema. Discriminator — current evidence associates routine morphine with respiratory depression, hypotension, and worse outcomes; it is used cautiously/selectively, not as automatic first-line therapy.

7. O2 + noninvasive ventilation

WHAT YOU SEE

The pink 'O2 and noninvasive ventilation' box on the right of the 'Pulmonary edema' band — the central intervention for the hypoxic, severely dyspneic patient.

WHAT IT ENCODES

Oxygen and NONINVASIVE positive-pressure ventilation (CPAP or BiPAP) are central to hypoxic pulmonary edema: positive intrathoracic pressure recruits alveoli, improves oxygenation/work of breathing, reduces preload AND afterload, and reduces the need for intubation. Give supplemental O2 only for hypoxemia (target SpO2 ≥90%) — routine O2 in non-hypoxemic patients is not beneficial. NIV is contraindicated in obtundation, vomiting, or inability to protect the airway.

BOARD TRAP

WRONG: give high-flow supplemental oxygen to every acute HF patient to be safe. Discriminator — supplemental O2 is for HYPOXEMIA (SpO2 <90%); hyperoxia in non-hypoxemic patients causes coronary/systemic vasoconstriction and is not beneficial. NIV (not blanket O2) is the key intervention for the hypoxic, dyspneic patient.

8. Low output phenotype

WHAT YOU SEE

Third band: hypoperfusion with end-organ dysfunction

WHAT IT ENCODES

The 'low output' phenotype is the COLD patient — hypoperfusion with end-organ dysfunction: narrow pulse pressure (proportional pulse pressure <25% suggests low cardiac index), cool/mottled extremities, fatigue, somnolence, cardio-renal syndrome (rising creatinine), and hepatic congestion (transaminitis, congestive hepatopathy). It is usually advanced systolic HF and carries a worse prognosis; perfusion must be restored before/alongside decongestion.

BOARD TRAP

WRONG: rising creatinine in a low-output HF patient always means over-diuresis, so stop diuretics and give fluids. Discriminator — in the cold/low-output patient, worsening renal function is often from POOR FORWARD PERFUSION and venous congestion (cardio-renal syndrome); improving cardiac output (inotropes if hypotensive) and careful decongestion — not IV fluid boluses — is the fix.

9. Inotropes if low BP/refractory

WHAT YOU SEE

The pink 'Inotropic therapy (if low blood pressure or diuretic refractoriness)' box on the right of the third 'Low output' band — the parenthetical IS the only indication.

WHAT IT ENCODES

In the low-output phenotype, inotropes (dobutamine/milrinone) are added ONLY for low blood pressure with hypoperfusion OR diuretic refractoriness (to raise cardiac output and renal perfusion so diuretics can work). They are a bridge — to recovery, transplant, or mechanical support — not a chronic outpatient therapy, given their O2-demand, arrhythmia, and mortality liabilities.

BOARD TRAP

WRONG: every low-output ADHF patient should be started on an inotrope. Discriminator — inotropes are reserved for low-output patients who are HYPOTENSIVE/hypoperfused or DIURETIC-REFRACTORY; a low-output patient with acceptable BP responding to diuretics does not need them, and routine use worsens outcomes.

10. Hemodynamic monitoring

WHAT YOU SEE

The pink 'Hemodynamic monitoring (suboptimal initial therapeutic response)' box centered in the 'Low output' band — the parenthetical flags it as a rescue tool for non-responders, not routine.

WHAT IT ENCODES

Invasive hemodynamic monitoring (pulmonary artery/Swan-Ganz catheter) is NOT routine — the ESCAPE trial showed no benefit from routine PAC-guided therapy in ADHF. It is reserved for patients with a SUBOPTIMAL response to initial therapy, diagnostic uncertainty about volume/perfusion status, cardiogenic shock, or evaluation for advanced therapies (transplant/MCS), where filling pressures and cardiac output directly guide titration.

BOARD TRAP

WRONG: a pulmonary artery (Swan-Ganz) catheter should be placed in all ADHF admissions to guide diuresis. Discriminator — routine PA catheterization did NOT improve outcomes (ESCAPE) and adds risk; reserve it for refractory/uncertain cases, shock, or advanced-therapy evaluation.

11. Cardiogenic shock phenotype

WHAT YOU SEE

Fourth band: hypotension, low CO, end-organ failure

WHAT IT ENCODES

Cardiogenic shock is the extreme: sustained HYPOTENSION (SBP <90 for >30 min or vasopressors needed) with a low cardiac index (<~2.2 L/min/m²), adequate/high filling pressures, plus end-organ hypoperfusion (oliguria, cold clammy skin, altered mentation, lactate >2). Most often due to acute MI with large LV infarct (mechanical complications — papillary muscle rupture, VSD, free-wall rupture — must be excluded). Mortality is high; rapid revascularization for MI-shock and escalation of support are key.

BOARD TRAP

WRONG: cardiogenic shock is managed primarily with aggressive IV diuresis to relieve congestion. Discriminator — diuretics alone do NOT treat shock; the patient is dying of inadequate cardiac OUTPUT/perfusion and needs inotropes/vasopressors, mechanical circulatory support, and treatment of the cause (urgent revascularization in MI-shock).

12. Inotropes (catecholamines)

WHAT YOU SEE

Inotropic therapy box (shock)

WHAT IT ENCODES

In cardiogenic shock, support perfusion with inotropes — usually CATECHOLAMINES. Norepinephrine is the preferred first-line vasopressor (the SOAP II subgroup and a dedicated trial showed fewer arrhythmias and lower mortality than dopamine in cardiogenic shock); add dobutamine for inotropy. Use the minimum dose to maintain a MAP ~65 and organ perfusion while arranging definitive therapy (revascularization, MCS).

BOARD TRAP

WRONG: dopamine is the first-line pressor for cardiogenic shock. Discriminator — NOREPINEPHRINE is preferred over dopamine in cardiogenic shock (dopamine caused more arrhythmias and higher mortality in SOAP II); reach for norepinephrine ± dobutamine, not dopamine.

13. Mechanical circulatory support

WHAT YOU SEE

MCS box: IABP, pVAD, ultrafiltration

WHAT IT ENCODES

When drugs fail in cardiogenic shock, escalate to MECHANICAL CIRCULATORY SUPPORT: intra-aortic balloon pump (IABP — note the IABP-SHOCK II trial showed NO mortality benefit in MI-shock, so it is no longer routinely first-choice), percutaneous VADs (Impella, TandemHeart), or VA-ECMO for biventricular/refractory shock — as a bridge to recovery, decision, transplant, or durable LVAD. Ultrafiltration (mechanical fluid removal) is an option for diuretic-resistant volume overload but is not superior to stepped diuretics for routine decongestion (CARRESS-HF).

BOARD TRAP

WRONG: an intra-aortic balloon pump improves survival and should be placed routinely in MI cardiogenic shock. Discriminator — IABP did NOT reduce mortality in MI cardiogenic shock (IABP-SHOCK II) and is no longer routine; escalate to active percutaneous VADs (Impella) or VA-ECMO for true refractory shock.

14. Wet/dry × warm/cold profiles

WHAT YOU SEE

NEW: 2x2 hemodynamic profile framework underlying the figure

WHAT IT ENCODES

The unifying bedside framework behind every phenotype is the 2x2 of CONGESTION (wet vs dry) and PERFUSION (warm vs cold). WET = congestion signs (orthopnea, elevated JVP, edema, rales, S3); COLD = hypoperfusion (narrow pulse pressure, cool extremities, rising creatinine, altered mentation). The four profiles: WET-WARM (most common — diuretics ± vasodilators), WET-COLD (decongest + inotropes/support — the sickest), DRY-WARM (compensated — optimize oral GDMT), DRY-COLD (hypovolemic/end-stage — cautious fluids/inotropes). Classifying the patient into one quadrant directly selects therapy.

BOARD TRAP

WRONG: ADHF therapy is one-size-fits-all — diurese everyone hard. Discriminator — you must first place the patient in a wet/dry × warm/cold quadrant: only WET patients need decongestion, only COLD patients need inotropes/support, and a DRY-COLD patient may actually need cautious FLUIDS; treating all comers with aggressive diuresis harms the dry or hypoperfused patient.